



EARN TO FIRE

Answer STEM questions. Fire the arena. Battle the bots.



Think

Your team answers robot science, maths and coding questions on an iPad.



Fire

Get one right and a real arena hazard fires: the flipper, the pit or the spinners.



Fight

Your drivers battle bots in the arena. Keep yours out of the pit!

Two teams. Two bots each. Two minutes.

Your group splits into **Red** and **Blue**. Two of each team drive the bots; the rest are the brains on the iPad. Your score is **STEM points + bot points**, so the brains and the drivers both win it for you.

What you'll learn

 **Forces & levers:** how a flipper lifts a robot

 **Code & logic:** loops, binary, IF statements

 **Circuits & electricity:** volts, amps, Ohm's law

 **Sensors & robot brains:** how robots sense the world

 **Gears & motion:** speed, RPM, gear ratios

 **Boss challenge:** energy, momentum and more

Got one wrong? The iPad explains the right answer, so you learn something either way.

How does the arena work?

Each iPad talks over WiFi to a small server. The server sends the command to a **Raspberry Pi**, a credit-card-sized computer, which switches the motors and the pneumatics. **Look through the clear front of the arena base to see the tech!**

⚠ Safety: the arena hazards only move when our referee has **armed** it, and they switch themselves off after a few seconds. Never reach into the arena. Bots go in and out only when the referee says so.



SPLIT THE GROUP

"You're two teams: **Red** and **Blue**. In each team, **two of you drive a bot**. Everyone else is the **brains** on your team's iPad."

THE BRAINS

"Tap a weapon and answer the question. **Get it right and that weapon fires for real**, and you score its points. Get it wrong and the iPad tells you why. Then try a different weapon."

THE DRIVERS

"Keep your bots **out of the pit**. You score a point every second each bot survives. Push the other team's bots in and your team gets **50 points** for each one!"

TEAMWORK TIP

"Brains: watch where their bots are. **Fire the hazards when their bots are close!**"

SAFETY

"**Hands stay out of the arena**. If I say **STOP**, everyone stops."

GO

"Two minutes on the clock. Highest score wins. **Ready? Go!**"

Group of 10: ● **Red** 5 (2 drivers + 3 brains) vs □ **Blue** 5 (2 drivers + 3 brains). Odd numbers: the extra joins the brains. Small group? One driver per team can drive both bots in turn.



Tap a weapon on the iPad and answer its question.

Right answer = it FIRES and you score.

WEAPON	QUESTION TOPIC	POINTS	RECHARGE
 Flipper	Forces & levers	10	8 s
 The Pit	Circuits & electricity	15	20 s
 Spinner 1	Gears & motion	20	15 s
 Spinner 2	Code & logic	20	15 s
 Spinner 3	Sensors & robot brains	20	15 s
 MEGA SPIN	Boss challenge (all three spinners!)	50	30 s

**Bot points**

+1 every second, for each of your bots still in

+50 for each of the other team's bots pitted

Remember

Wrong answer? Read why, then wait 15 s before that weapon again. Try another meanwhile!

Recharge is shared. Fire it first and the other team has to wait.

YOUR SCORE = STEM POINTS + BOT POINTS



The 7-minute slot

0:00	Group arrives. New round (everything) . Split into Red / Blue
0:30	Read the briefing (page 2). Drivers take controllers
1:30	Hands out. ARM. START the clock
3:30	Round ends. DISARM . Read out the Standings
4:00	Winner! Leaderboard line. Teacher handout (page 6)
5:00	Bots back to start. Next group in

~50 groups a day. Longer slot? Play a second round and swap drivers and brains.

Start of each day

- ☐ Team iPads: <http://192.168.0.102/kids/>, pick Red / Blue
- ☐ Referee iPad: </kids/admin.html>, PIN from the lead (never written down)
- ☐ Shows **SAFE**; both iPads say **PRACTICE MODE**
- ☐ One practice answer per weapon: scores move, arena doesn't
- ☐ Arena clear. **ARM**, fire each weapon, watch it stop
- ☐ **ALL STOP**: everything stops, back to SAFE
- ☐ Bot batteries charged; spares on charge

⚠ Safety rules

- **DISARM before any hand goes near the arena.** Every time.
- **ALL STOP** = disarm + stop clock + all spinners off. Use it freely.
- Spinners stop themselves after 5 s; the pit comes back up after 10 s.
- If the server drops out mid-spin, a spinner may keep going. **Switch off arena power.**
- After any restart the arena comes back **SAFE**.

During the round

Tap **PITTED** under a bot the moment it drops. Mis-tap? **Undo**. Both of a team's bots pitted = **KNOCKOUT**, the round ends.

Buttons

SAFE / ARMED	SAFE = practice: answers score, nothing moves
START / STOP	Clock. STOP pauses the questions too
RESET	Clock back to 2:00, fresh fight
±5 / ±10	Fix STEM points by hand
Clear cooldowns	Unstick the weapons

If something goes wrong

<i>Cannot reach the arena</i>	Check iPad WiFi, reload page
Right answer, nothing moved	ARMED? Check log for <i>Pi FAILED</i>
Referee page shows PIN screen	Server restarted; log in (it's SAFE)
5 wrong PINs	Locked 60 s; wait, retry

EARN TO FIRE

Leaderboard · Day: _____ · Sheet ____

Top team score today

Best STEM score today

[illegible]

Red / Blue = team totals from the referee iPad's **Standings** (STEM + bot points). KO = both of one team's bots pitted. 20 groups per sheet, so **print 3 per day** (9 for the event).

Your class played **Earn to Fire** with InfoSec Battle Bots at STEMfest Newcastle. In two teams, students answered STEM questions to fire real hazards in a robot combat arena while classmates drove the robots. Every wrong answer came with an explanation.

Topics covered (KS3)

Forces & levers	Fulcrums, Newton's laws, weight = mass × g, centre of gravity
Circuits & electricity	Ohm's law ($V = I \times R$), $P = V \times I$, series circuits, fuses, short circuits
Gears & motion	Speed = distance / time, RPM, gear ratios, circumference
Code & logic	Variables, loops, IF statements, binary, AND / NOT
Sensors & robot brains	Ultrasonic ranging, inputs vs outputs, actuators, microcontrollers

Back in class: five follow-up questions

1. A spinner weapon spins **twice as fast**. Why does it hit **four times** harder? ($KE = \frac{1}{2}mv^2$)
2. A 12 V battery drives a 4 Ω motor. What current flows, and how much power does it use?
3. Design a pit-avoiding robot: which sensor would you use, and what would your **IF** statement say?
4. Why do battle robots use a **low centre of gravity**? Sketch a robot that is hard to flip.
5. Count to 31 in binary on one hand. How high can you count with two hands?

Answers: (1) energy grows with speed squared, so $2^2 = 4$; (2) 3 A, 36 W; (3) e.g. a downward light or ultrasonic sensor: IF floor not detected THEN reverse; (4) it must be lifted further before it tips; (5) 1023 (10 bits).

How the arena works

The iPads talk over WiFi to a Windows server running a Node.js program. It checks each answer, then sends commands to a **Raspberry Pi**, which switches the motors and pneumatics. It is real networking, coding and electronics: a good talking point for careers in engineering and cyber security.

Bring InfoSec Battle Bots to your school

Our schools programme is led by **Lewis Terry**. For school visits, workshops and follow-up resources, get in touch:

Email	info@infosecbattlebots.com (mark it "Schools")
Schools page	www.infosecbattlebots.com/schools
Follow us	@InfosecBots on X
The game's code	github.com/Ben-PyroGuy-Doch/ibb-arena (open source)